1. Difference between letter of transmittal and authorization 3 marks

Letter of Transmittal:

This element is included in relatively formal and very formal reports. Its purpose is to release or deliver the report to the recipient. It also serves to establish some rapport between the reader and the writer. This is one part of the formal report where a personal, or even a slightly informal, tone should be used. The transmittal letter should not dive into report findings except in the broadest terms. This letter may be like:

Letter of Authorization:

This is a letter to the researcher approving the project, detailing who has responsibility for the project and indicating what resources are available to support it. The letter not only shows who sponsored the research but also delineates the original request.

Researcher would not write this letter. In many situations, referring to the letter of authorization in the letter of transmittal is sufficient. If so, the letter of authorization need not be included in the report. In case the letter has to be included, exact copy of the original may be reproduced.

Qno. 1 Difference between population and target population.

Population

A population is the theoretically specified aggregation of study elements. It is translating the abstract concept into workable concept. For example, let us look at the study of "college students." Theoretically who are the college students? They might include students registered in government colleges and/or private colleges, students of intermediate classes and/or graduate classes, students of professional colleges and/or non-professional colleges, and many other variations. In this way the pool of all available elements is population.

Target Population

Out of the conceptual variations what exactly the researcher wants to focus on. This may also be called a target population. Target population is the complete group of specific population elements relevant to the research project. Target population may also be called survey population i.e. that aggregation of elements from which the survey sample is actually selected.

At the outset of the sampling process, it is vitally important to carefully define the target population so that the proper source from which the data is to be collected can be identified. In our example of „college students" finally we may decide to study the college students from government institutions located in Lahore, who are studying social sciences, who are aged 19 years of age, and hailing from rural areas.

Three important Parts of experiment

Parts of Experiments:

We can divide the experiments into seven parts and for each part there is a term. Not all experiments have all these parts, and some have all seven parts plus others. The following seven usually make up a true experiment.

1. Treatment or independent variable.
2. Dependent variable.
3. Pretest.
4. Posttest.
5. Experimental group.
6. Control group.
7. Assignment of subjects.

Qno. 1 why pilot testing is important before questionnaire survey? 5

Pilot testing also called pre-testing means small scale trial run of a particular component; here we are referring to pilot testing of the questionnaire. Conventional wisdom suggests that pre-testing not only is an established practice for discovering errors but also is useful for extra training the research team. Ironically, professionals who have participated in scores of studies are more likely to pretest an instrument than is a beginning researcher hurrying to complete a project. Revising questions five or more times is not unusual. Yet inexperienced researchers often underestimate the need to follow the design-test-revise process.

It is **important to pilot test the instrument** to ensure that the questions are understood by the respondents and there are no problems with the wording or measurement. Pilot testing involves the use of a small number of respondents to test the appropriateness of the questions and their comprehension. Usually, the draft questionnaire is tried out on a group that is selected on a convenience and that is similar in makeup to the one that ultimately will be sampled. Making a mistake with 25 or so subjects can avert the disaster of administering an invalid questionnaire to several hundred individuals. Hence the main purpose of pilot testing is to identify potential problems with the methods, logistics, and the questionnaire.

Administering a questionnaire exactly as planned in the actual study often is not possible. For example, mailing out a questionnaire might require several weeks. Pre-testing a questionnaire in this manner might provide important information on response rate, but it may not point out why questions were skipped or why respondents found certain questions ambiguous or confusing. The ability of personal interviewer to record requests for additional explanation and to register comments indicating respondent‟s difficulty with question sequence or other factors is the primary reason why interviewers are often used for pretest work.

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5 duties of field researcher.5

A field researcher does the following:

- Observes ordinary events and everyday activities as they happen in natural settings, in addition to unusual occurrences.
- Becomes directly involved with people being studied and personally experiences the process of daily life in the field setting.
- Acquires an insider‟s point of view while maintaining the analytic perspective or distance of an outsider.
- Uses a variety of techniques and social skills in a flexible manner as the situation demands.
- Produces data in the form of extensive, written notes, as well as diagrams, maps, pictures to provide very detailed descriptions.
- Sees events holistically (as a whole, not in pieces) and individually in their social context.
- Understands and develops empathy for members in a field setting, and does not just record „cold‟ objective facts.
- Notices both explicit (recognized, conscious, spoken) and tacit (less recognized, implicit, unspoken) aspects of culture.
- Observes ongoing social processes without upsetting, or imposing an outside point of view.
- Copes with high levels of personal stress, uncertainty, ethical dilemmas, and ambiguity.

Steps in Field Research

Naturalism and direct involvement mean that field research is more flexible or less structured than quantitative research. This makes it essential for a researcher to be well organized and prepared for the field. It also means that the steps of project are not entirely predetermined but serve as an approximate guide or road map. Here is just the listing of these steps:

1. Prepare yourself, read the literature and defocus.
2. Select a site and gain access.
3. Enter the field and establish social relations with members.
4. Adopt a social role, learn the ropes, and get along with members.
5. Watch, listen, and collect quality data.
6. Begin to analyze data, generate and evaluate working hypothesis.
7. Focus on specific aspects of the setting and use theoretical sampling.
8. Conduct field interviews with member informants.
9. Disengage and physically leave the setting.
10. Complete the analysis and write the report.

Convenience sampling with e.g

Convenience sampling (also called haphazard or accidental sampling) refers to sampling by obtaining units or people who are most conveniently available. For example, it may be convenient and economical to sample employees in companies in a nearby area, sample from a pool of friends and neighbors. The person-on-the street interview conducted by TV programs is another example. TV interviewers go on the street with camera and microphone to talk to few people who are convenient to interview. The people walking past a TV studio in the middle of the day do not represent everyone (homemakers, people in the rural areas). Likewise, TV interviewers select people who look “normal” to them and avoid people who are unattractive, poor, very old, or inarticulate.

Another **example** of haphazard sample is that of a newspaper that asks the readers to clip a questionnaire from the paper and mail it in. Not everyone reads the newspaper, has an interest in the topic, or will take the time to cut out the questionnaire, and mail it. Some will, and the number who does so may seem large, but the sample cannot be used to generalize accurately to the population.

Convenience samples are least reliable but normally the cheapest and easiest to conduct. Convenience sampling is most often used during the exploratory phase of a research project and is perhaps the best way of getting some basic information quickly and efficiently. Often such sample is taken to test ideas or even to gain ideas about a subject of interest.

Data collection method

1. **Data collection methods:** Did the data come from primary sources or secondary sources? How the primary data were collected – survey, experiment, observation? It is possible that multiple techniques may have been used – all these have to be explained.

Parts of table

1. Give each table a number.
2. Give each table a title, which names variables and provides background information
3. Label the row and columns variables and give name to each of the variable categories.
4. Include the totals of the columns and rows. These are called marginal. They equal the univariate frequency distribution for the variable.
5. Each number or place that corresponds to the intersection of a category for each variable is a cell of a table.
6. The numbers with the labeled variable categories and the totals are called the body of the table.
7. If there is missing information, report the number of missing cases near the table to account for all original cases.

Researchers convert raw count tables into percentages to see bivariate relationship. There are three ways to percentage a table: by row, by column, and for the total. The first two are often used and show relationship.

Is it best to percentage by row or column? Either could be appropriate. A researcher‟s hypothesis may imply looking at row percentages or the column percentages. Here, the hypothesis is that age affects attitude, so column percentages are most helpful. Whenever one factor in a cross-tabulation can be considered the cause of the other, percentage will be most illuminating if they are computed in the direction of the causal factor.

Which dilemma a person feel with deviant

Involvement with deviants: Researchers who conduct research on deviants who engage in illegal behavior face additional dilemmas. They know of and are sometimes involved in illegal activity. They might be getting „guilty knowledge.‟ Such knowledge is of interest not only to law enforcement officials but also to other deviants. The researcher faces a dilemma of building trust and rapport with the deviants, yet not becoming so involved as to violate his or her basic personal moral standards. Usually, the researcher makes an explicit arrangement with the deviant members.

1.Survey interview and field interview difference 1.5 marks and why standard of question remain same in survey interview and changed in field interview? 1.5 marks

Survey Interview	Field Interview
1. It has clear beginning and end.	1. The beginning and end are not clear. The interview can be picked up later.
2. The same standard questions are asked of all respondents in the same sequence.	2. The questions and the order in which they are asked are tailored to specific people and situations.
3. The interviewer appears neutral at all times.	3.The interviewer shows interest in responses, encourages elaboration.
4. The interviewer asks questions, and the respondent answers.	4. It is like a friendly conversational exchange but with more interviewer questions.
5. It is almost always with one respondent alone.	5. It can occur in group setting or with others in area, but varies.
6. It has a professional tone and businesslike focus, diversions are ignored.	6. It is interspersed with jokes, aside, stories, diversions, and anecdotes, which are recorded.
7. Closed-ended questions are common, with rare probes.	7. Open-ended questions are common, and probes are frequent.
8. The interviewer alone controls the pace and direction of interview.	8. The interviewer and member jointly control the pace and direction of the interview.
9. The social context in which the interview occurs is ignored and assumed to make little difference.	9. The social context of the interview is noted and seen as important for interpreting the meaning of responses.
10. The interviewer attempts to mold the communication pattern into a standard framework.	10. The interviewer adjusts to the member‟s norms and language usages.

1- Explain importance of statistical software (MINITAB and SPSS) in data presentation. Give examples

The proliferation of computer technology in business and universities has greatly facilitated tabulation and statistical analysis. Commercial packages eliminate the need to write a new program every time you want to tabulate and analyze data with a computer. SAS, Statistical Package for the Social Sciences (**SPSS**), SYSTAT, Epi, Info, and **MINITAB** is commonly used statistical packages. These user friendly packages emphasize statistical calculations and hypothesis testing for varied types of data. They also provide programs for entering and editing data. Most of these packages contain sizeable arrays of programs for descriptive analysis and univariate, bivariate, and multivariate statistical analysis.

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disadvantages of content analysis

Disadvantages

1. **Bias:** Many documents used in research were not originally intended for research purposes. The various goals and purposes for which documents are written can bias them in various ways. For example, personal documents such as confessional articles or autobiographies are often written by famous people or people who had some unusual experience such as having been a witness to a specific event. While often providing a unique and valuable research data, these documents usually are written for the purpose of making money. Thus they tend to exaggerate and even fabricate to make good story. They also tend to include those events that make the author look good and exclude those that cast him or her in a negative light.
2. **Selective survival:** Since documents are usually written on paper, they do not withstand the elements well unless care is taken to preserve them. Thus while documents written by famous people are likely to be preserved, day-to-day documents such as letters and diaries written by common people tend either to be destroyed or to be placed in storage and thus become inaccessible. It is relatively rare for common documents that are not about some events of immediate interest to the researcher (e.g., suicide) and not about famous occurrence or by some famous person to be gathered together in a public repository that is accessible to researchers.
3. **Incompleteness:** Many documents provide incomplete account to the researcher who has had no prior experience with or knowledge of the events or behavior discussed. A problem with many personal documents such as letters and diaries is that they were not written for research purposes but were designed to be private or even secret. Both these kinds of documents often assume specific knowledge that researcher unfamiliar with certain events will not possess. Diaries are probably the worst in this respect, since they are usually written to be read only by the author and can consist more of "soul searching" and confession than of description. Letters tend to be little more complete, since they are addressed to a second person. Since many letters assume a great amount of prior information on the part of the reader.
4. **Lack of availability of documents:** In addition to the bias, incompleteness, and selective survival of documents, there are many areas of study for which no documents are available. In many cases information simply was never recorded. In other cases it was recorded, but the documents remain secret or classified, or have been destroyed.
5. **Sampling bias:** One of the problems of bias occurs because persons of lower educational or income levels are less likely to be represented in the sampling frames. The problem of sampling bias by educational level is more acute for document study than for survey research. It is a safe generalization that a poorly educated people are much less likely than well educated people to write documents.
6. **Limited to verbal behavior:** By definition, documents provide information only about respondent‟s verbal behavior, and provide no direct information on the respondent‟s nonverbal behavior, either that of the document‟s author or other characters in the document.
7. **Lack of standardized format:** Documents differ quite widely in regard to their standardization of format. Some documents such as newspapers appear frequently in a standard format. Large dailies always contain such standard components as editorial page, business page, sports page, and weather report. Standardization facilitates comparison across time for the same newspapers and comparison across different newspapers at one point in time. However, many other documents, particularly personal documents have no standard format. Comparison is difficult or impossible, since valuable information contained in the document at one point in time may be entirely lacking in an earlier or later documents.
8. **Coding difficulties:** For a number of reasons, including differences in purpose for which the documents were written, differences in content or subject matter, lack of standardization, and differences in length and format, coding is one of the most difficult tasks facing the content analyst. Documents are generally written arrangements, rather than numbers are quite difficult to quantify. Thus analysis of documents is similar to analysis of open-ended survey questions.
9. **Data must be adjusted for comparability over time:** Although one of the advantages of document study is that comparisons may be made over a long period of time, since external events cause changes so drastic that even if a common unit of measure is used for the entire period, the value of this unit may have changed so much over time that comparisons are misleading unless corrections are made. Look at the change in measuring distance, temperature, currency, and even literacy in Pakistan.

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with help of examples explain why we prefer probability sampling over non probability sampling

There are several alternative ways of taking a sample. The major alternative sampling plans may be grouped into probability techniques and non-probability techniques. In **probability sampling** every element in the population has a known nonzero probability of selection. The simple random is the best known probability sample, in which each member of the population has an equal probability of being selected. Probability sampling designs are used when the representativeness of the sample is of importance in the interest of wider generalizability. When time or other factors, rather than generalizability, become critical, non-probability sampling is generally used.

In non-probability sampling the probability of any particular element of the population being chosen is unknown. The selection of units in non-probability sampling is quite arbitrary, as researchers rely heavily on personal judgment. It should be noted that there are no appropriate statistical techniques for measuring random sampling error from a non-probability sample. Thus projecting the data beyond the sample is statistically inappropriate. Nevertheless, there are occasions when non-probability samples are best suited for the researcher"s purpose.

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uses of experimental design

Experimental design provides the **theory and methods to be applied in selecting** the test design, which includes the choice of test scenarios and their time sequence; determination of the sample size; and the use of techniques such as randomization, controls, matching or blocking, and sequential testing.

Important because **it allows us to estimate the inherent variability in the data**. This allows us to judge whether an observed difference could be due to chance variation. treatment factors are variables in the experiment, within the control of the experimenter that we think could affect the response.

Define Mean, Median and Mode

Researchers often want to summarize the information about one variable into a single number. They use three measures of central tendency, or measures of the center of the frequency distribution:

mean, median and mode,

mean which are often called averages (a less precise and less clear way to say the same thing).

The mode is simply the most common or frequently occurring number.

The median is the middle point.

The mean also called the arithmetic average, is the most widely used measure of central tendency. A particular central tendency is used depending upon the nature of the data.

According to scenario give advantages of Focus Group Discussion FGD

- The primary advantage of focus groups is its ability to quickly and inexpensively grasp the core issues of the topic. One might see focus group discussions as synergistic i.e. the combined effort of the group will produce a wider range of information, insights, and ideas than will the accumulation of separately secured responses of a number of individuals. Even in non-exploratory research, focus group discussions produce a lot more information far more quickly, and at less cost than individual interviews.
- As part of exploratory research, focus group discussions help the researcher to focus on the issue and develop relevant research hypotheses. In the discussions the relevant variables are identified, and relationships are postulated. Once the variables are identified, the same focus group discussions help in the formulation of questions, along with the response categories, for the measurement of variables.
- Focus group discussion is an excellent design to get information from non-literates.
- Focus groups discussions are a good means to discover attitudes and opinions that might not be revealed through surveys. This is particularly useful when the researcher is looking at the controversial issue, and the individual might be able to give his opinion as such but not discuss the issue in the light of other viewpoints. In focus group discussions there is usually a snowballing effect. A comment by one often triggers a chain of views from other participants.
- Focus group discussions are well accepted in the folk communities, as this form of communication already exists whereby the local communities try to sort out controversial issues.
- Focus group discussions generate new ideas, questions about the issues under consideration. It may be called serendipity (surprise ideas). It is more often the case in a group than in an individual interview that some idea will drop out of the blue. The group also affords the opportunity to develop the idea to its full significance.
- Focus group discussions can supplement the quantitative information on community knowledge, attitude, and practice (KAP), which may have already been collected through survey research.

- Focus group discussions are highly flexible with respect to topic, number of participants, time schedule, location, and logistics of discussion.
- Focus group discussions provide a direct link between the researcher and the population under study. In fact most of the focus group discussions are held close to people‟s places of living and work. It helps in getting the realistic picture of the issue directly from the people who are part of it.
- For some researchers, focus group discussions may be a fun. They enjoy discussing the issues directly with the relevant population.

- The prefatory part of the research report has six components. Enlist three.

Prefatory parts

1. Title fly page
2. Title page
3. Letter of transmittal
4. Letter of authorization
5. Table of contents
6. Executive summary

Title Fly Page: Only the title appears on this page. For the most formal reports, a title fly page precedes the title page. Most of the reports don‟t have it. May be it is more like the dustcover of some books.

Title Page: The title page should include four items: the title of the report, the name(s) of the person(s) for whom the report was prepared, the name(s) of person(s) who prepared it, and the date of release or presentation.

The title should be brief but include three elements: (1) the variables included in the study, (2) the type of relationship among the variables, and (3) the population to which the results may be applied. Redundancies such as "A report of," "A discussion of," and "A study of" add length to title but little else. Single-word titles are also of little value.

Addresses and titles of recipients and writers may also be included.

(For thesis follow the format as prescribed by the relevant university)

Letter of Transmittal: This element is included in relatively formal and very formal reports. Its purpose is to release or deliver the report to the recipient. It also serves to establish some rapport between the reader and the writer. This is one part of the formal report where a personal, or even a slightly informal, tone should be used. The transmittal letter should not dive into report findings except in the broadest terms. This letter may be like:

3.Why researchers take sample instead of study the whole population?

Sampling is the process of using a small number of items or parts of a larger population to make conclusions about the whole population. It enables the researchers to estimate unknown characteristics of the population.

What you analyze from Solomon group design explain it.

To gain more confidence in internal validity in experimental designs, it is advisable to set up two experimental groups and two control groups. One experimental group and one control group can be given the both pretest and the posttest. The other two groups will be given only the posttest. Here the effects of treatment can be calculated in several different ways as shown in figure 1:

Group	Pretest	Treatment	Posttest
1. Experimental	O1	X	O2

2. Control	O3	-	O4
3. Experimental	-	X	O5
4. Control	-	-	O6

(O2 – O1) = E

(O4 - O3) = E

(O5 - O6) = E

(O5 - O3) = E

[(O2 - O1) – (O4 – O3)] = E

E = Effect

1.Correlational research

Correlational research is **a type of non-experimental research method in which a researcher measures two variables, understands and assesses the statistical relationship between them with no influence from** any extraneous variable.

2.Simple and Random techniques

The simple random sample is both the easiest random sample to understand and the one on which other types are modeled. In simple random sampling, a research develops an accurate sampling frame, selects elements from sampling frame according to mathematically random procedure, then locates the exact element that was selected for inclusion in the sample.

After numbering all elements in a sampling frame, the researcher uses a list of random numbers to decide which elements to select. He or she needs as many random numbers as there are elements to be sampled: for example, for a sample of 100, 100 random numbers are needed. The researcher can get random numbers from a random number table, a table of numbers chosen in a mathematically random way. Random-number tables are available in most statistics and research methods books. The numbers are generated by a pure random process so that any number has an equal probability of appearing in any position. Computer programs can also produce lists of random number.

4.Explain pre- test and post-test with example

Frequently a researcher measures the dependent variable more than once during an experiment. The pretest is the measurement of the dependent variable prior to the introduction of the treatment. The posttest is the measurement of the dependent variable after the treatment has been introduced into the experimental situation.

8.What are the limitations of trivariate table.

Trivariate tables have three limitations. First, they are difficult to interpret if a control variable has more that four categories. Second, control variables can be at any level of measurement, but interval or ratio control variables must be grouped (i.e. converted to an ordinal level), and how cases are grouped can affect the interpretation of effects. Finally, the total number of cases is a limiting factor because the cases are divided among cells in partials. The number of cells in the partials equals the number of cells in the bivariate relationship multiplied by the number of categories in the control variables. For example if the control variable has three categories, and a bivariate table has 12 cells, the partials have 3 X 12 = 36 cells. An average of five cases per cell is recommended, so the researcher will need 5 X 36 = 180 cases at minimum.

Like a bivariate table construction, a trivariate table begins with a compound frequency distribution (CFD), but it is a three-way instead of two-way CFD. An example of a trivariate table with “gender” as control variable for the bivariate table is shown here:

9.Why random sampling technique are better than other technique.

Random samples are most likely to yield a sample that truly represents the population. In addition, random sampling lets a researcher statistically calculate the relationship between the sample and the population – that is the size of sampling error. A non-statistical definition of the sampling error is the deviation between sample result and a population parameter due to random process.

Random samples are the best method of selecting your sample from the population of interest. The advantages are that your **sample should represent the target population and eliminate sampling bias**. The disadvantage is that it is very difficult to achieve (i.e. time, effort and money).

10.Types of errors in instrument

- Gross Errors.
- Random Errors.
- Systematic Errors.

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11. Ethnography and Ethno methodology

Two modern extensions of field research, ethnography and ethno-methodology, build on the social constructionist perspective.

Ethnography comes from cultural anthropology. Ethno means people or a folk distinct by their culture and ethnography refers to describing something. Thus ethnography means describing a culture and understanding another way of life from the native point of view. It is just an understanding the culture of people from their own perspective.

Ethno-methodology implies how people create reality and how they interpret it. Ethno-

methodologists examine ordinary social interaction in great detail to identify the rules for constructing social reality and common sense, how these rules are applied, and how new rules are created. They try to figure out how certain meanings are attached to a reality.

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13. how can experimental design be used in Social sciences

Experimental research builds on the principles of positivist approach more directly than do the other research techniques. Researchers in the natural sciences (e.g. chemistry and physics), related applied fields (e.g. engineering, agriculture, and medicines) and the social sciences conduct experiments. The logic that guides an experiment on plant growth in biology or testing a metal in engineering is applied in experiments on human social behavior. Although it is most widely used in psychology, the experiment is found in education, criminal justice, journalism, marketing, nursing, political science, social work, and sociology.

14.Dilemma of field research

Ethical Dilemmas of Field research

The direct personal involvement of a field researcher in the social lives of other people raises many ethical dilemmas. The dilemmas arise when the researcher is alone in the field and has little time to make a moral decision. Although he or she may be aware of general ethical issues before entering the field, they arise unexpectedly in the course of observing and interacting in the field. Let us look at some of these dilemmas:

Deception: Deception arises in several ways in field research. The research may be covert; or may assume a false role, name, or identity; or may mislead members in some way. The most hotly debated of the ethical issues arising from deception is that of covert versus overt field research. Some support it and see it as necessary for entering into and aiming a full knowledge of many areas of social life. Others oppose it and argue that it undermines a trust between researchers and society. Although its moral status is questionable, there are some field sites or activities that can only be studied covertly. One may have to look into the cost and benefit equation; where the researcher is the best judge.

Covert research is never preferable and never easier than overt research because of the difficulties of maintaining a front and the constant fear of getting caught.

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17. Reactive or non-reactive 5

Experiments and surveys research are both **reactive**; that is, the people being studied are aware of the fact that they are being studied. In **non-reactive** research, those being studied are not aware that they are part of a research project. Such a research is largely based on positivistic principles but is also used by interpretive and critical researchers.

The Logic of Non-Reactive Research

The critical thing about non-reactive or unobtrusive measures (i.e. the measures that are not obtrusive or intrusive) is that the people being studied are not aware of it but leave evidence of their social behavior or actions naturally. The researcher infers from the evidence to behavior or attitudes without disrupting those being studied. Unnoticed observation is also a type of non-reactive measure. For example, a researcher may be observing the behavior of drivers from a distance whether drivers stopped at red sign of the traffic lights. The observations can be made both at the day time and at night. It could also be noted whether the driver was a male or a female; whether the driver was alone or with passengers; whether other traffic was present; and whether the car came to a complete stop, a slow stop, or no stop.

18.Statistic or parameter Ki definition

Parameter

A parameter is the summary description of a given variable in a population. The mean income of all families in a city and the age distribution of the city‟s population are parameters. An important part of survey research involves the estimation of population parameters on the basis of sample observation.

Statistic

A statistic is the summary description of a given variable in a survey sample. Thus the mean income computed from the survey sample and the age distribution of that sample are statistics. Sample statistics are used to make estimates of the population parameters.

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conditions of causal relationships. Explain 3 marks

A basic causal relationship requires only independent and dependent variable. A third type of variable, the intervening variable, appears in more complex causal relationships. It comes between the independent and dependent variables and shows the link or mechanism between them. Advances in knowledge depend not only on documenting cause and effect relationship but also on specifying the mechanisms that account for the causal relation. In a sense, the intervening variable acts as a dependent variable with respect to independent variable and acts as an independent variable toward the dependent variable.

3.what are central tendency which method is mostly use explain.

Researchers often want to summarize the information about one variable into a single number. They use three measures of central tendency, or measures of the center of the frequency distribution: mean, median and mode, which are often called averages (a less precise and less clear way to say the same thing). The mode is simply the most common or frequently occurring number. The median is the middle point. The mean also called the arithmetic average, is the most widely used measure of central tendency. A particular central tendency is used depending upon the nature of the data.

Non probability and convenience

In non-probability sampling designs, the elements in the population do not have any probabilities attached to their being chosen as sample subjects. This means that the findings from the study of the sample cannot be confidently generalized to the population. However the researchers may at times be less concerned about generalizability than obtaining some preliminary information in a quick and inexpensive way. Sometimes non-probability could be the only way to collect the data.

Convenience Sampling

Convenience sampling (also called haphazard or accidental sampling) refers to sampling by obtaining units or people who are most conveniently available. For example, it may be convenient and economical to sample employees in companies in a nearby area, sample from a pool of friends and neighbors. The person-on-the street interview conducted by TV programs is another example. TV interviewers go on the street with camera and microphone to talk to few people who are convenient to interview. The people walking past a TV studio in the middle of the day do not represent everyone (homemakers, people in the rural areas). Likewise, TV interviewers select people who look "normal" to them and avoid people who are unattractive, poor, very old, or inarticulate.

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Target population and sampling frame

Target Population

Out of the conceptual variations what exactly the researcher wants to focus on. This may also be called a target population. Target population is the complete group of specific population elements relevant to the research project. Target population may also be called survey population i.e. that aggregation of elements from which the survey sample is actually selected.

At the outset of the sampling process, it is vitally important to carefully define the target population so that the proper source from which the data is to be collected can be identified. In our example of „college students" finally we may decide to study the college students from government institutions located in Lahore, who are studying social sciences, who are aged 19 years of age, and hailing from rural areas.

Sampling

Sampling is the process of using a small number of items or parts of a larger population to make conclusions about the whole population. It enables the researchers to estimate unknown characteristics of the population.

Sampling Frame

In actual practice the sample will be drawn from a list of population elements that is often different from the target population that has been defined. A sampling frame is the list of elements from which the sample may be drawn. A simple example could be listing of all college students meeting the criteria of target population and who are enrolled on the specified date.

A sampling frame is also called the working population because it provides the list that can be worked with operationally. In our example, such a list could be prepared with help of the staff of the selected colleges.

2.measures of central tendency for skewed data 3 marks

The median is the most informative measure of central tendency for skewed distributions or distributions with outliers. For example, the median is often used as a measure of central tendency for income distributions, which are generally highly skewed.

Criteria for selecting sampling technique

Depending upon the type of topic, the researcher lays down the criteria for the subjects to be included in the sample. Whoever meets that criteria could be selected in the sample? The researcher might select such cases or might provide the criteria to somebody else and leave it to his/her judgment for the actual selection of the subjects. That is why such a sample is also called as **judgmental or expert opinion sample.** For example a researcher is interested in studying students who are enrolled in a course on research methods, are highly regular, are frequent participants in the class discussions, and often come with new ideas. The criteria has been laid down, the researcher may do this job himself/herself, or may ask the teacher of this class to select the students by using the said criteria. In the latter situation we are leaving it to the judgment of the teacher to select the subjects. Similarly we can give some criteria to the fieldworkers and leave it to their judgment to select the subjects accordingly. In a study of working women the researcher may lay down the criteria like: the lady is married, has two children, one of her child is school going age, and is living in nuclear family.